02 Computer Arithmetic 200Q

Topic 2 Computer Arithmetic & Digital Logic - 200 MCQs

Syllabus: Propositional/Predicate Logic, WFF, Satisfiability/Tautology, Logic Families TTL/ECL/CMOS, Boolean algebra & minimization (K-map, Quine-McCluskey), Flip-flops, race, combinational/sequential design, Number systems (Octal/Hex/Decimal/Binary), 1's/2's complement arithmetic, Floating point (IEEE 754). Distribution: ~50 logic, ~100 digital circuits, ~50 arithmetic.

Total Questions: 200 | Level: PG + UGC-NET + Faculty Screening | Mix: Theory, Numerical, Application

---

Q1. Which of the following is a tautology?

A. p ? ¬p

Q2. Which formula is unsatisfiable (contradiction)?

A. p ? ¬p

Q3. A wff is satisfiable iff?

A. It is fals

Q4. De Morgan's law states ¬(p ? q) equals?

A. p ? q

Q5. p ? q is logically equivalent to?

A. ¬q ? p

Q6. Which logic family has the highest speed?

A. TTL

Q7. Which logic family has lowest static power dissipation?

A. CMOS

Q8. TTL stands for?

A. Transis

Q9. ECL's key characteristic is?

A. Saturat

Q10. CMOS inverter in steady state draws?

A. Near-ze

Q11. Predicate ?x P(x) is true when?

A. P holds

Q12. ?x ?y R(x,y) versus ?y ?x R(x,y): the former?

A. Is alway

Q13. Skolemization is used to?

A. Minimiz

Q14. A valid argument's conclusion must be true when?

A. At least

Q15. Resolution principle applies to formulas in?

A. Horn fo

Q16. Which of the following is logically equivalent to ¬(p ? q)?

A. p ? q

Q17. The dual of Boolean expression A·(B+C) is?

A. A+B·C

Q18. Fan-out of a logic gate refers to?

A. Power

Q19. Noise margin is highest (among classic families) in?

A. CMOS

Q20. A tautology with n variables is true in how many rows of truth table?

A. 4-1 row

Q21. Which of the following is logically equivalent to ¬(p ? q)?

A. ¬p ? ¬q

Q22. The dual of Boolean expression A·(B+C) is?

A. (A+B)·C

Q23. Fan-out of a logic gate refers to?

A. Numbe

Q24. Noise margin is highest (among classic families) in?

A. ECL

Q25. A tautology with n variables is true in how many rows of truth table?

A. 16-1 ro

Q26. Which of the following is logically equivalent to ¬(p ? q)?

A. ¬p ? ¬q

Q27. The dual of Boolean expression A·(B+C) is?

A. A+B·C

Q28. Fan-out of a logic gate refers to?

A. Numbe

Q29. Noise margin is highest (among classic families) in?

A. ECL

Q30. A tautology with n variables is true in how many rows of truth table?

A. All 16 r

Q31. Which of the following is logically equivalent to ¬(p ? q)?

A. p ? q

Q32. The dual of Boolean expression A·(B+C) is?

A. A+B·C

Q33. Fan-out of a logic gate refers to?

A. Propag

Q34. Noise margin is highest (among classic families) in?

A. I2L

Q35. A tautology with n variables is true in how many rows of truth table?

A. 2 rows

Q36. Which of the following is logically equivalent to ¬(p ? q)?

A. ¬p ? ¬q

Q37. The dual of Boolean expression A·(B+C) is?

A. A+B·C

Q38. Fan-out of a logic gate refers to?

A. Numbe

Q39. Noise margin is highest (among classic families) in?

A. CMOS

Q40. A tautology with n variables is true in how many rows of truth table?

A. All 4 ro

Q41. Which of the following is logically equivalent to ¬(p ? q)?

A. p ? q

Q42. The dual of Boolean expression A·(B+C) is?

A. A·B+C

Q43. Fan-out of a logic gate refers to?

A. Propag

Q44. Noise margin is highest (among classic families) in?

A. ECL

Q45. A tautology with n variables is true in how many rows of truth table?

A. 2 rows

Q46. Which of the following is logically equivalent to ¬(p ? q)?

A. p ? q

Q47. The dual of Boolean expression A·(B+C) is?

A. A+(B·C

Q48. Fan-out of a logic gate refers to?

A. Power

Q49. Noise margin is highest (among classic families) in?

A. Schottk

Q50. A tautology with n variables is true in how many rows of truth table?

A. All 16 r

Q51. An n-variable K-map (n=4) has how many cells?

A. 8

Q52. In a K-map, groups must contain cells in powers of?

A. 2 (1,2,4

Q53. Quine-McCluskey method is used for?

A. Floating

Q54. An essential prime implicant is one that?

A. Covers

Q55. A 4:1 multiplexer has how many select lines?

A. 4

Q56. A 4-to-16 decoder has how many output lines?

A. 16

Q57. Race-around condition occurs in?

A. D flip-fl

Q58. Master-slave JK flip-flop was introduced to overcome?

A. Fan-in

Q59. Characteristic equation of D flip-flop (next state Q+) is?

A. Q+ = J

Q60. Characteristic equation of JK flip-flop is?

A. Q+ = T

Q61. A mod-8 ripple counter requires at least how many flip-flops?

A. 8

Q62. A combinational circuit's output depends on?

A. Presen

Q63. A sequential circuit's output depends on?

A. Tempe

Q64. Gray code of binary 1011 is? (adjacent codes differ by 1 bit)

A. 1010

Q65. The Boolean function F=A+A'B equals (absorption)?

A. A

Q66. Minimum number of NAND gates to implement NOT, AND, OR respectively (using NAND only)?

A. 1,1,1

Q67. A half-adder has inputs/outputs?

A. 2 inputs

Q68. A full-adder can be built from how many half-adders + OR?

A. 1 half-a

Q69. Propagation delay in ripple-carry adder grows as?

A. O(n) wi

Q70. Edge-triggered flip-flop samples input on?

A. Entire H

Q71. An n-variable K-map (n=3) has how many cells?

A. 8

Q72. In a K-map, groups must contain cells in powers of?

A. 3

Q73. Quine-McCluskey method is used for?

A. Encodi

Q74. An essential prime implicant is one that?

A. Covers

Q75. A 2:1 multiplexer has how many select lines?

A. 2

Q76. A 4-to-16 decoder has how many output lines?

A. 5

Q77. Race-around condition occurs in?

A. ROM

Q78. Master-slave JK flip-flop was introduced to overcome?

A. Race-a

Q79. Characteristic equation of D flip-flop (next state Q+) is?

A. Q+ = D

Q80. Characteristic equation of JK flip-flop is?

A. Q+ = J

Q81. A mod-10 ripple counter requires at least how many flip-flops?

A. 5

Q82. A combinational circuit's output depends on?

A. Presen

Q83. A sequential circuit's output depends on?

A. Tempe

Q84. Gray code of binary 1011 is? (adjacent codes differ by 1 bit)

A. 1110

Q85. The Boolean function F=A+A'B equals (absorption)?

A. A'B

Q86. Minimum number of NAND gates to implement NOT, AND, OR respectively (using NAND only)?

A. 2,2,2

Q87. A half-adder has inputs/outputs?

A. 2 inputs

Q88. A full-adder can be built from how many half-adders + OR?

A. 2 half-a

Q89. Propagation delay in ripple-carry adder grows as?

A. O(log n

Q90. Edge-triggered flip-flop samples input on?

A. Entire H

Q91. An n-variable K-map (n=2) has how many cells?

A. 4

Q92. In a K-map, groups must contain cells in powers of?

A. 2 (1,2,4

Q93. Quine-McCluskey method is used for?

A. Designi

Q94. An essential prime implicant is one that?

A. Covers

Q95. A 2:1 multiplexer has how many select lines?

A. 2

Q96. A 3-to-8 decoder has how many output lines?

A. 8

Q97. Race-around condition occurs in?

A. JK flip-f

Q98. Master-slave JK flip-flop was introduced to overcome?

A. Clock s

Q99. Characteristic equation of D flip-flop (next state Q+) is?

A. Q+ = Q

Q100. Characteristic equation of JK flip-flop is?

A. Q+ = S

Q101. A mod-16 ripple counter requires at least how many flip-flops?

A. 5

Q102. A combinational circuit's output depends on?

A. Clock c

Q103. A sequential circuit's output depends on?

A. Power s

Q104. Gray code of binary 1011 is? (adjacent codes differ by 1 bit)

A. 0100

Q105. The Boolean function F=A+A'B equals (absorption)?

A. A'B

Q106. Minimum number of NAND gates to implement NOT, AND, OR respectively (using NAND only)?

A. 3,2,1

Q107. A half-adder has inputs/outputs?

A. 3 inputs

Q108. A full-adder can be built from how many half-adders + OR?

A. 4 half-a

Q109. Propagation delay in ripple-carry adder grows as?

A. O(log n

Q110. Edge-triggered flip-flop samples input on?

A. Entire L

Q111. An n-variable K-map (n=4) has how many cells?

A. 16

Q112. In a K-map, groups must contain cells in powers of?

A. 3

Q113. Quine-McCluskey method is used for?

A. Floating

Q114. An essential prime implicant is one that?

A. Is alway

Q115. A 2:1 multiplexer has how many select lines?

A. 2

Q116. A 4-to-16 decoder has how many output lines?

A. 8

Q117. Race-around condition occurs in?

A. D flip-fl

Q118. Master-slave JK flip-flop was introduced to overcome?

A. Power

Q119. Characteristic equation of D flip-flop (next state Q+) is?

A. Q+ = Q

Q120. Characteristic equation of JK flip-flop is?

A. Q+ = D

Q121. A mod-8 ripple counter requires at least how many flip-flops?

A. 8

Q122. A combinational circuit's output depends on?

A. Presen

Q123. A sequential circuit's output depends on?

A. Presen

Q124. Gray code of binary 1011 is? (adjacent codes differ by 1 bit)

A. 1010

Q125. The Boolean function F=A+A'B equals (absorption)?

A. A

Q126. Minimum number of NAND gates to implement NOT, AND, OR respectively (using NAND only)?

A. 2,2,2

Q127. A half-adder has inputs/outputs?

A. 3 inputs

Q128. A full-adder can be built from how many half-adders + OR?

A. 3 half-a

Q129. Propagation delay in ripple-carry adder grows as?

A. O(log n

Q130. Edge-triggered flip-flop samples input on?

A. Active

Q131. An n-variable K-map (n=3) has how many cells?

A. 8

Q132. In a K-map, groups must contain cells in powers of?

A. 3

Q133. Quine-McCluskey method is used for?

A. Designi

Q134. An essential prime implicant is one that?

A. Is alway

Q135. A 2:1 multiplexer has how many select lines?

A. 1

Q136. A 3-to-8 decoder has how many output lines?

A. 8

Q137. Race-around condition occurs in?

A. JK flip-f

Q138. Master-slave JK flip-flop was introduced to overcome?

A. Fan-in

Q139. Characteristic equation of D flip-flop (next state Q+) is?

A. Q+ = J

Q140. Characteristic equation of JK flip-flop is?

A. Q+ = S

Q141. A mod-8 ripple counter requires at least how many flip-flops?

A. 8

Q142. A combinational circuit's output depends on?

A. Previou

Q143. A sequential circuit's output depends on?

A. Presen

Q144. Gray code of binary 1011 is? (adjacent codes differ by 1 bit)

A. 1110

Q145. The Boolean function F=A+A'B equals (absorption)?

A. A

Q146. Minimum number of NAND gates to implement NOT, AND, OR respectively (using NAND only)?

A. 2,2,2

Q147. A half-adder has inputs/outputs?

A. 2 inputs

Q148. A full-adder can be built from how many half-adders + OR?

A. 4 half-a

Q149. Propagation delay in ripple-carry adder grows as?

A. O(n) wi

Q150. Edge-triggered flip-flop samples input on?

A. Both le

Q151. Decimal 109 in 8-bit binary is?

A. 110110

Q152. Decimal 222 in hexadecimal (2 digits) is?

A. 0xDF

Q153. Decimal 86 in octal is?

A. 127

Q154. Range of signed integers in 3-bit 2's complement is?

A. -4 to +3

Q155. 1's complement of binary 01101 is?

A. 01101

Q156. In IEEE 754 single precision, exponent bias is?

A. 1023

Q157. In IEEE 754 single precision (32-bit), bit-fields are?

A. 1+7+24

Q158. In IEEE 754 double precision, exponent bias is?

A. 1023

Q159. Binary 1100 + 0111 equals decimal?

A. 18

Q160. Overflow in 2's complement addition occurs when?

A. There i

Q161. Decimal 103 in 8-bit binary is?

A. 011001

Q162. Decimal 55 in hexadecimal (2 digits) is?

A. 0x38

Q163. Decimal 87 in octal is?

A. 126

Q164. Range of signed integers in 5-bit 2's complement is?

A. -31 to 3

Q165. 1's complement of binary 10110 is?

A. 01001

Q166. In IEEE 754 single precision, exponent bias is?

A. 128

Q167. In IEEE 754 single precision (32-bit), bit-fields are?

A. 1 sign +

Q168. In IEEE 754 double precision, exponent bias is?

A. 1023

Q169. Binary 1110 + 0010 equals decimal?

A. 17

Q170. Overflow in 2's complement addition occurs when?

A. Result i

Q171. Decimal 14 in 8-bit binary is?

A. 000111

Q172. Decimal 194 in hexadecimal (2 digits) is?

A. 0xC2

Q173. Decimal 80 in octal is?

A. 117

Q174. Range of signed integers in 7-bit 2's complement is?

A. 0 to 12

Q175. 1's complement of binary 10011 is?

A. 11111

Q176. In IEEE 754 single precision, exponent bias is?

A. 127

Q177. In IEEE 754 single precision (32-bit), bit-fields are?

A. 1 sign -

Q178. In IEEE 754 double precision, exponent bias is?

A. 63

Q179. Binary 0010 + 0011 equals decimal?

A. 6

Q180. Overflow in 2's complement addition occurs when?

A. Operan

Q181. Decimal 79 in 8-bit binary is?

A. 010011

Q182. Decimal 73 in hexadecimal (2 digits) is?

A. 0x4A

Q183. Decimal 38 in octal is?

A. 26

Q184. Range of signed integers in 6-bit 2's complement is?

A. -32 to +

Q185. 1's complement of binary 01101 is?

A. 00000

Q186. In IEEE 754 single precision, exponent bias is?

A. 1023

Q187. In IEEE 754 single precision (32-bit), bit-fields are?

A. 1+7+24

Q188. In IEEE 754 double precision, exponent bias is?

A. 127

Q189. Binary 0011 + 0111 equals decimal?

A. 21

Q190. Overflow in 2's complement addition occurs when?

A. Result i

Q191. Decimal 145 in 8-bit binary is?

A. 100100

Q192. Decimal 25 in hexadecimal (2 digits) is?

A. 0x19

Q193. Decimal 18 in octal is?

A. 23

Q194. Range of signed integers in 7-bit 2's complement is?

A. 0 to 12

Q195. 1's complement of binary 11001 is?

A. 11111

Q196. In IEEE 754 single precision, exponent bias is?

A. 1023

Q197. In IEEE 754 single precision (32-bit), bit-fields are?

A. 1+11+2

Q198. In IEEE 754 double precision, exponent bias is?

A. 2047

Q199. Binary 0110 + 0110 equals decimal?

A. 12

Q200. Overflow in 2's complement addition occurs when?

A. Carry i

ANSWER KEY

1-A, 2-C, 3-C, 4-B, 5-C, 6-B, 7-A, 8-A, 9-D, 10-A

11-B, 12-B